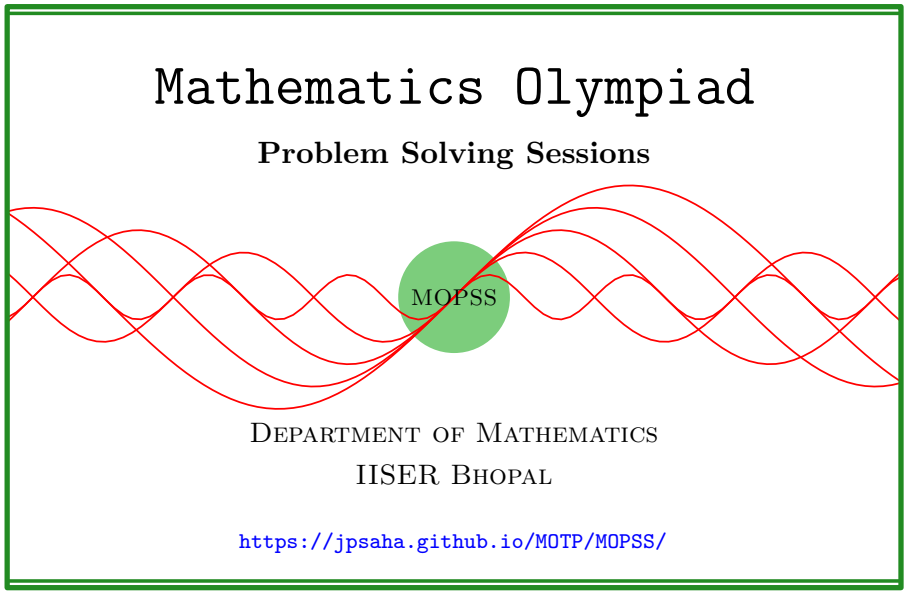
Power of a point



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (RMO 2004 P1,) 2
2 Exercise (RMO 2015f P4) 2

§1 Power of a point and radical axis

Exercise 1 (RMO 2004 P1,). Consider in the plane a circle Γ with center O and a line ℓ not intersecting circle Γ. Prove that there is a point Q on the perpendicular drawn from O to the line ℓ, such that for any point P on the line ℓ, PQ represents the length of the tangent from P to the circle Γ.

Exercise 2 (RMO 2015f P4). Two circle Σ_1 and Σ_2 having centres C_1 and C_2 intersect at A and B. Let P be a point on the segment AB and let $AP \neq BP$. The line through P perpendicular to C_1P meets Σ_1 at C and D. The line through P perpendicular to C_2P meets Σ_2 at E and F. Prove that C, D, E, F are the vertices of a rectangle.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)